

ATIVIDADE DE APOIO – ENSINO MÉDIO

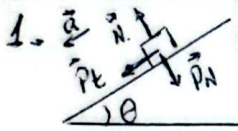
Componente Curricular: Física

Professor (a): Jazana

Assunto (s): Plano Inclinado

Data: Outubro

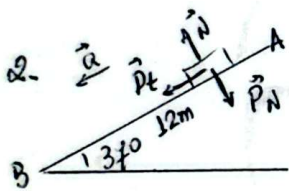
Série: 1ª



$m = 2 \text{ kg}$
 $N = P_N$
 $N = 20 \cdot 0,8 \Rightarrow N = 16 \text{ N}$

$\text{1º e 2º e 3º): } P_t = P \cdot \sin \theta$
 $P_t = 20 \cdot 0,6$
 $P_t = 12 \text{ N}$
 $P_t = m \cdot a$
 $12 = 2 \cdot a$
 $a = \frac{12}{2} \Rightarrow a = 6 \text{ m/s}^2$

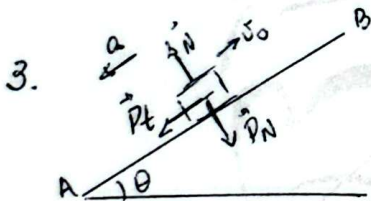
ou $\text{2º e 3º): } a = g \cdot \sin \theta$
 $a = 10 \cdot 0,6$
 $a = 6 \text{ m/s}^2$



$m = 1 \text{ kg}$
 $v_0 = 0$

a) $P_t = P \cdot \sin 30^\circ$
 $P_t = 10 \cdot 0,6$
 $P_t = 6 \text{ N}$

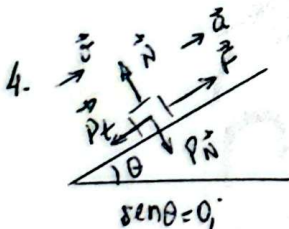
b) $P_t = m \cdot a$
 $6 = 1 \cdot a$
 $a = 6 \text{ m/s}^2$
 $\Delta S = v_0 t + \frac{a t^2}{2}$
 $12 = \frac{6 \cdot t^2}{2}$
 $t^2 = 4 \Rightarrow t = 2 \text{ s}$



a) $a = g \cdot \sin \theta$
 $a = 10 \cdot 0,3$
 $a = 3 \text{ m/s}^2$

b) $v = v_0 + at$
 $0 = 12 - 3t$
 $3t = 12$
 $t = \frac{12}{3} \Rightarrow t = 4 \text{ s}$

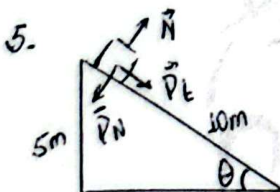
c) $v^2 = v_0^2 + 2a \Delta S$
 $0 = 12^2 + 2 \cdot (-3) \cdot \Delta S$
 $6 \Delta S = 144$
 $\Delta S = \frac{144}{6} \Rightarrow \Delta S = 24 \text{ m} \therefore d = 18 \text{ m}$
 $d = 24 \text{ m}$



$m = 2 \text{ kg}$
 $F = 22 \text{ N}$

$P_t = P \cdot \sin \theta$
 $P_t = 20 \cdot 0,7$
 $P_t = 14 \text{ N}$

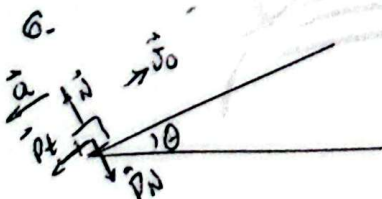
$F_R = m \cdot a$
 $F - P_t = m \cdot a$
 $22 - 14 = 2 \cdot a \Rightarrow 8 = 2 \cdot a \Rightarrow a = 4 \text{ m/s}^2$



$\sin \theta = \frac{5}{10}$
 $\sin \theta = \frac{1}{2}$

a) $a = g \cdot \sin \theta$
 $a = 10 \cdot \frac{1}{2} \Rightarrow a = 5 \text{ m/s}^2$

$\Delta S = v_0 t + \frac{a t^2}{2}$
 $10 = \frac{5 \cdot t^2}{2}$
 $t^2 = 4 \Rightarrow t = 2 \text{ s}$



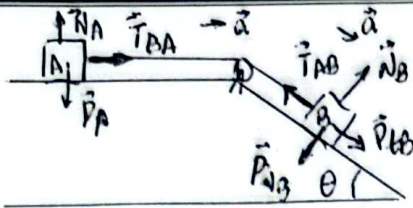
$v = 0$

$a = g \cdot \sin \theta$

$v^2 = v_0^2 + 2a \Delta S$
 $0 = v_0^2 + 2 \cdot (-g \sin \theta) \cdot \Delta S$
 $2g \sin \theta \cdot \Delta S = v_0^2$
 $\Delta S = \frac{v_0^2}{2g \sin \theta}$

$d = 18 \text{ m}$
 $d = \frac{v_0^2}{2g \sin \theta}$

4-



* $T_{BA} = T_{AB}$

$m_A = 2 \text{ kg}$

$m_B = 3 \text{ kg}$

* $P_{tB} = P_B \sin \theta$

$P_{tB} = 30 \cdot 0,5$

$P_{tB} = 15 \text{ N}$

B $\Rightarrow P_{tB} - T_{AB} = m_B a$

A $\Rightarrow T_{BA} = m_A \cdot a$

$P_{tB} = (m_A + m_B) \cdot a$

$15 = (2 + 3) \cdot a$

$15 = 5 \cdot a \Rightarrow a = \frac{15}{5} \Rightarrow a = 3 \text{ m/s}^2$

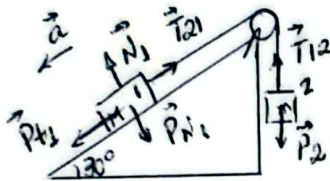
* $T_{BA} = m_A \cdot a$

$T_{BA} = 2 \cdot 3$

$T_{BA} = 6 \text{ N}$

||

8-



* $T_{12} = T_{21}$

$a = 2 \text{ m/s}^2$

* $P_{t1} = P_1 \sin \theta$

$P_{t1} = 10 \text{ M} \cdot \frac{1}{2} \Rightarrow P_{t1} = 5 \text{ M}$

* 1 $\Rightarrow P_{t1} - T_{21} = M \cdot 2$

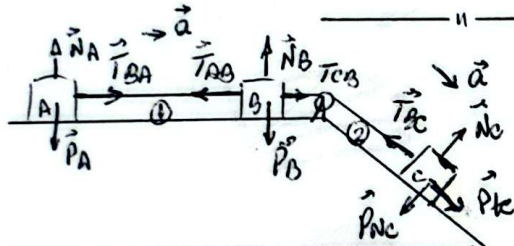
$T_{12} - P_2 = m \cdot 2$

$P_{t1} - P_2 = 2M + 2m$

$5M - 10m = 2M + 2m$

$3M = 12m \Rightarrow \frac{M}{m} = \frac{12}{3} \Rightarrow \frac{M}{m} = 4$

9-



* $P_{tC} = P_C \sin \theta$

$P_{tC} = m \cdot g \cdot \frac{1}{2} \Rightarrow P_{tC} = \frac{mg}{2}$

* $T_{BA} = T_{AB}$

* $T_{BC} = T_{CB}$

$m_A = 2m$

$m_B = 2m$

$m_C = m$

a) * C $\Rightarrow P_{tC} - T_{BC} = m_C \cdot a$

B $\Rightarrow T_{CB} - T_{AB} = m_B \cdot a$

A $\Rightarrow T_{BA} = m_A \cdot a$

$P_{tC} = (m_A + m_B + m_C) \cdot a$

$\frac{mg}{2} = (2m + 2m + m) \cdot a$

$\frac{mg}{2} = 5m \cdot a \Rightarrow a = \frac{mg}{10}$

$a = \frac{1}{5} g \Rightarrow a = \frac{g}{5}$

b) * $T_{BA} = m_A \cdot a$

$T_{BA} = 2m \cdot \frac{g}{10} \Rightarrow T_{BA} = \frac{mg}{5}$

* $P_{tC} - T_{BC} = m_C \cdot a$

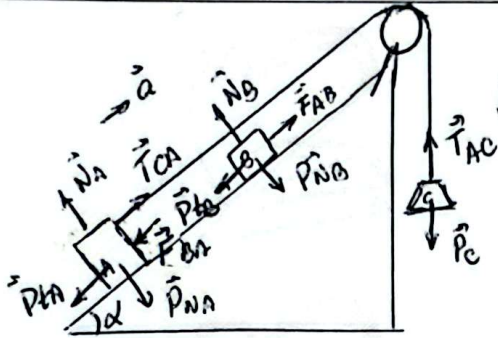
$0,5mg - T_{BC} = m \cdot \frac{g}{10} \Rightarrow T_{BC} = \frac{1}{2} mg - \frac{mg}{10}$

$T_{BC} = \frac{5mg - mg}{10} = \frac{4mg}{10} \Rightarrow T_{BC} = \frac{2mg}{5}$

c) Sistema em repouso $\Rightarrow$ Cada bloco em repouso $\Rightarrow \vec{r} = 0$

$F = P_{tC}$
 $F = \frac{mg}{2}$

10.



$$* T_{CA} = T_{AC}$$

$$* F_{AB} = F_{BA}$$

$$\begin{aligned} m_A &= 6 \text{ kg} \\ m_B &= 4 \text{ kg} \\ m_C &= 10 \text{ kg} \end{aligned}$$

$$\begin{aligned} * P_{tA} &= P_A \cdot \sin \alpha \\ P_{tA} &= 60 \cdot 0,6 \\ P_{tA} &= 36 \text{ N} \end{aligned}$$

$$\begin{aligned} * P_{tB} &= P_B \cdot \sin \alpha \\ P_{tB} &= 40 \cdot 0,6 \\ P_{tB} &= 24 \text{ N} \end{aligned}$$

$$* C \rightarrow P_C - T_{AC} = m_C \cdot a$$

$$* P_C = m_C \cdot g$$

$$P_C = 10 \cdot 10 = 100 \text{ N}$$

$$B \rightarrow F_{AB} - P_{tB} = m_B \cdot a$$

$$A \rightarrow T_{CA} - P_{tA} - F_{BA} = m_A \cdot a$$

$$P_C - P_{tA} - P_{tB} = (m_A + m_B + m_C) \cdot a$$

$$100 - 36 - 24 = (6 + 4 + 10) \cdot a$$

$$40 = 20 \cdot a$$

$$a = \frac{40}{20} \Rightarrow a = 2 \text{ m/s}^2$$

$$* F_{AB} - P_{tB} = m_B \cdot a$$

$$F_{AB} - 24 = 4 \cdot 2$$

$$F_{AB} = 8 + 24 \Rightarrow \boxed{F_{AB} = 32 \text{ N}}$$